Business Glossary – Entity Resolution System

Conceptual framing

This system distinguishes carefully between **real-world entities** (the referents that exist independently of the system), **representations** (information artefacts that describe or refer to those entities), and **identifiers** (symbols or signs used to reference system-managed artefacts). The system does not store or manipulate real-world entities directly; instead, it manages identity through **clusters of entity mentions**, which embody the system's current **conceptualisation** of co-reference under available evidence.

A cluster scaffolds a **canonical entity** (a logical identity construct), and both are identified by a **canonical identifier**. This canonical identifier functions as a **proxy reference** to a real-world entity via the cluster it identifies. Canonical entity representations, when present, are optional descriptive artefacts that may be derived, curated, replaced, or omitted without changing identity.

I. Conceptual & Information-Oriented Glossary

This section defines domain concepts, identity constructs, information artefacts, and data-level elements. These terms describe **what exists conceptually** and **what the system manages as information**, independent of concrete system components or runtime behaviour.

Preferred term	Definition	Scope and use context	Alternative terms
Real-World Entity	A concrete or abstract object that exists independently of the system and is the intended target of reference.	A real-world entity is a referent. It is *not* stored, versioned, or directly identified by the system. All system-managed artefacts (mentions, clusters, identifiers) only ever refer to real-world entities indirectly and provisionally.	referent, real-world object, real entity, domain entity, business object (conceptual)

Entity Mention	A record-level information artefact submitted to the system by a source system/actor that describes or refers to a real-world entity.	An entity mention includes an Entity Mention Representation (payload + format) and request context (e.g. originator, timestamps, declared entity type). They are representations, not identities. They are the atomic inputs to entity resolution and may be duplicated, inconsistent, incomplete, or wrong. They can be reprocessed multiple times (e.g. during rebuilds).	entity record, mention, input entity, source record, submitted entity
Mention Identifier	A system-scoped identifier assigned to an entity mention as a stored information artefact.	Identifies an information artefact (i.e the entity mention), not a real-world entity. It must remain stable for traceability but carries no identity semantics beyond the mention itself.	record identifier, mention identifier
Entity Mention Representation	The descriptive payload of an entity mention, expressed in a specific serialised form.	Supplied externally as part of a mention submission. Used for transport, parsing, validation, and extraction. It is not an identity construct. It includes both the payload and the indication of its format (e.g. JSON, XML, RDF).	mention payload, input payload, source payload
Entity Representation	A descriptive information artefact that conveys information about an entity without asserting or carrying identity.	Entity Representation is an abstract concept realised concretely as either an Entity Mention Representation or a Canonical Entity Representation. It exists solely to carry descriptive content and must never be used as an identity construct, identifier, or evidence of equivalence. Representations	entity description, descriptive artefact, entity payload

		may be duplicated, inconsistent, incomplete, or replaced without affecting identity.	
Cluster	An operational identity construct identified by a canonical identifier and defined by the set of entity mention assignments that the system currently asserts to co-refer to the same real-world entity under available evidence.	A Cluster is the primary identity-bearing construct within the system. It embodies a hypothesis of co-reference, not a fact about the real world. Cluster composition may change over time due to new evidence, curation, or rebuild operations, while the canonical identifier remains stable whenever possible.	identity cluster, equivalence class, co-reference group
Canonical Entity	A stable, opaque, system-generated identifier assigned to a canonical entity (cluster) and used as the sole indirect reference to a real-world entity within ERS. The canonical identifier is deterministically derived and has no required protocol, dereferenceability, or RDF semantics.	A canonical entity is not a record, not a representation, and not a real-world entity. It exists only as the identity defined by a cluster and has no descriptive content unless a canonical representation is explicitly associated. It provides a stable identity anchor within the system.	resolved identity, identity construct, logical identity
Canonical Identifier	A stable, universal, system-generated identifier assigned to a cluster and its canonical entity.	Used across ERS APIs, Decision Store, Canonical Entity Registry, and external contracts to reference canonical entities. The canonical identifier is authoritative and monotonic: it is never revoked, even if assignments evolve.	canonical ID, stable identifier

Canonical Entity Representation	A derived or curated descriptive artefact associated with a canonical entity.	It is optional, non-authoritative representation. It must not be confused with a real-world entity or source record. It may be derived from mentions, curated by humans, or sourced authoritatively. It may change or disappear without affecting identity.	consolidated description, canonical description, derived entity description
Alignment (link)	An association between an entity mention and a canonical entity proposed by an automated engine or selected through curation. Alignments have no independent authority and are governed exclusively through Resolution Decisions.	Used within Resolution Decisions and decision context to describe candidate or selected associations. Alignments must not be interpreted as independently authoritative facts outside the decision framework.	match, linkage, correspondence, candidate link, proposed link
Default Canonical Alignment	The initial assignment of an entity mention to a canonical entity, established deterministically at intake. This alignment guarantees that every valid entity mention has a canonical identifier independent of automated resolution outcomes.	Used internally by ERS when registering new entity mentions and creating the initial Resolution Decision. Default canonical alignments may later be confirmed, replaced, or constrained by automated resolution, curation, or rebuild operations.	initial canonical assignment
Alignment Polarity	An explicit indication of whether an asserted correspondence between an entity mention and a canonical entity is positive (equivalence) or negative (non-equivalence).	Alignment Polarity allows the system to represent explicit rejection, not just absence of acceptance. It is essential for governance, rebuild constraints, and curator locks, ensuring that rejected correspondences are preserved as evidence and are not reintroduced by automated resolution or rebuild operations.	equivalence polarity, match sign, assertion polarity

Confidence Score	A numeric signal expressing either (a) engine-provided similarity assessment or (b) governance confidence applied to a resolution decision. These two uses are distinct and must not be conflated.	In ERS governance, confidence values are discrete (+1 confirmed, 0 provisional, −1 rejected) and authoritative. In ERE responses, confidence scores are continuous (0..1), advisory only, and used as input for decision-making.	matching score, assignment confidence,
Confidence Threshold	A configurable minimum confidence score required to accept an automatic alignment.	Governs workflow behaviour (auto-accept vs review) and precision/recall trade-offs.	matching threshold, acceptance threshold
Entity Type	An ontological category that classifies real-world entities according to their nature or role within a domain.	They define *what kind of thing* an entity is (e.g. Person, Organisation) and may constrain applicable properties, resolution rules, or matching strategies. Entity types are not tags or identifiers, they do *not* identify entities, and do *not* imply the existence of a canonical entity or cluster.	class, entity category, entity class, reference type, canonical entity type (CET)
Resolution Decision	A governance artefact that captures the current resolution outcome for a single entity mention, including candidate alignments, decision status, and the authoritative canonical assignment.	A Resolution Decision is the unit of governance in the system. It binds exactly one entity mention to one current canonical assignment, while retaining alternative candidate alignments and their evaluation context. Decisions are created by automated resolution, refined by human curation, and replaced during rebuild operations.	decision, resolution outcome, governance decision

II. System, Actors & Behaviour-Oriented Glossary

This section defines system components, actors, operations, and interaction patterns. These terms describe **how the system behaves**, **who performs actions**, and **where responsibilities lie**.

Preferred term	Definition	Scope and use context	Alternative terms
Originator	An external system or process that submits entity mentions or resolution requests.	The originator provides data but does not control resolution outcomes. Used for provenance and traceability.	source system
Entity Resolution	The process of assigning entity mentions to clusters that represent managed identities under available evidence.	Entity resolution operationally creates, updates, or revises cluster membership. It manages system identities and does not assert facts about real-world entities. An umbrella capability implemented jointly by ERS and ERE.	identity resolution
Entity Resolution Engine (ERE)	A processing system component responsible for executing resolution logic and maintaining operational cluster state.	Engines may be independently developed and replaced, provided they conform to the ER Engine Contract. They are pluggable to the ERsys. Internal data (of clusters, mentions, entities and their representations) is operational and non-authoritative.	ER engine, matching engine, resolution engine

Entity Resolution Service (ERS)	The coordinating system component responsible for APIs, orchestration, persistence of authoritative records (resolution inputs as a system of records and canonical entity registry), and interaction with EREs.	Acts as the authoritative system of record for entity mentions, resolution requests, and outcomes in a canonical entity registry. Does *not* perform resolution logic itself.	ER service, orchestration service, coordination service
Entity Resolution System (ERSys)	The complete information system responsible for resolving entity mentions into managed identities.	Encompasses the ERS, one or more EREs, link curation capabilities, persistence layers, and supporting infrastructure. It defines system boundaries for execution, governance, and accountability. It is *not* a single component or engine.	ER system, entity resolution platform, ER platform, ER solution
ER Engine Contract	A formal specification defining required interfaces, data structures, and behavioural guarantees for EREs.	Enables independent development, replacement, and parallel operation of multiple engines. It governs interoperability, not internal algorithms.	engine contract, engine interface, engine API contract
Automatic Alignment	An alignment produced automatically by an ERE.	Represents a proposed resolution decision, not a final one. Subject to thresholds and curation.	machine match
Default Canonical Alignment	A provisional assignment of an entity mention to a canonical entity, minted by the system in the	Used to guarantee that every valid entity mention has a canonical identifier within bounded time. Default assignments typically carry	provisional assignment, fallback canonical link

	absence of confident resolution evidence.	neutral or zero confidence and may later be refined, replaced, or confirmed by automated resolution, human curation, or rebuild operations.	
Manual Alignment	A human-in-the-loop process in which alignment decisions are explicitly accepted or rejected by a curator.	Link curation refines automated assertions for quality assurance and risk mitigation. Introduces expert judgement and accountability into the resolution lifecycle. This process refines or overrides automated assertions but does not bypass clustering rules or create identities directly.	Link Curation, curated match, human validation, human review
Alignment Status	The lifecycle state of an alignment assertion.	Used to distinguish proposed, accepted, rejected, or superseded assertions for audit, governance, and curation workflows. Assertion status is independent of confidence score.	decision state, assertion status
Curator	A human actor responsible for curating alignments.	Acts through the curation application; does not manipulate engine internals directly.	reviewer, data steward, domain expert, reviewer
Curator Lock	A manually asserted, authoritative constraint that confirms or rejects a specific alignment between an	Curator Locks are produced through human curation actions and expressed using normative confidence values (e.g. +1	manual lock, governed constraint, curator constraint

	entity mention and a canonical entity.	confirmed, −1 rejected). They are immutable constraints that must not be overridden by automated resolution or rebuild processing.	
Link Curation Web Application	A user-facing application supporting curation workflows.	Provides interfaces for review and decision-making; does not implement resolution logic.	curation UI
System of Records	An immutable store of all entity mentions and resolution requests submitted to the system, preserved with full provenance and submission context.	The System of Records is the authoritative source for what has been submitted, not for how resolution outcomes were produced. It contains entity mentions, their representations, and request metadata, and serves as the basis for replay, audit, and rebuild operations. It does not store resolution outcomes, decisions, or canonical state.	request log, input record store
Canonical Entity Registry	A persistent store that maintains the latest accepted canonical state for entity resolution, consisting of canonical identifiers and their current mention-to-canonical assignments.	The Canonical Entity Registry is a projection store, optimised for lookup and exposure of current canonical identity state. It intentionally retains no historical versions or decision rationale. Historical reasoning, alternative candidates, and governance context are held exclusively in the Decision Store. The registry may be rebuilt, refreshed, or replaced as resolution outcomes	resolved data repository, canonical registry, resolved entity registry

		evolve, without losing canonical identifier continuity.	
Decision Store	A persistent store maintaining the current resolution decision per entity mention, including candidate alignment context, decision status, and curator actions.	The Decision Store is the authoritative source of resolution traceability and governance context. It retains decision state and rationale, while the Canonical Entity Registry retains only the latest accepted canonical projection. The Decision Store is required for curation, rebuild, audit, and explainability.	decision repository, decision log, resolution decision store
Resolution Orchestration	Coordination logic managing the lifecycle of resolution requests and responses.	Bridges intake, engine invocation, persistence, and exposure via APIs.	resolution coordination
Rebuild	A resolution operation that recomputes clusters for all entity mentions in the system of records.	Implies reconstruction of the ERE's internal state from authoritative mention records.	global reclustering, re-execution, full rebuild, system-wide reclustering,
Re-resolve	A resolution operation that recomputes cluster membership for a specific entity mention subject to previously recorded rejection constraints. Re-resolution does not change the canonical identifier by itself.	Used within ERS governance flows (typically initiated from the Link Curation application) to request a new resolution attempt excluding one or more previously rejected canonical candidates. Re-resolution is distinct from lookup, replay, and rebuild.	re-resolution

Entity Type Configuration	A system configuration artefact defining how entity mention representations of a given type are interpreted and processed.	Specifies schema expectations and extractor instructions (e.g. XPath, JSONPath, SPARQL). It is *not* an ontological category; it is an operational configuration keyed by entity type. It may cover constraints on mention representation fields, resolution strategies, or matching logic.	entity type config, schema config,